NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

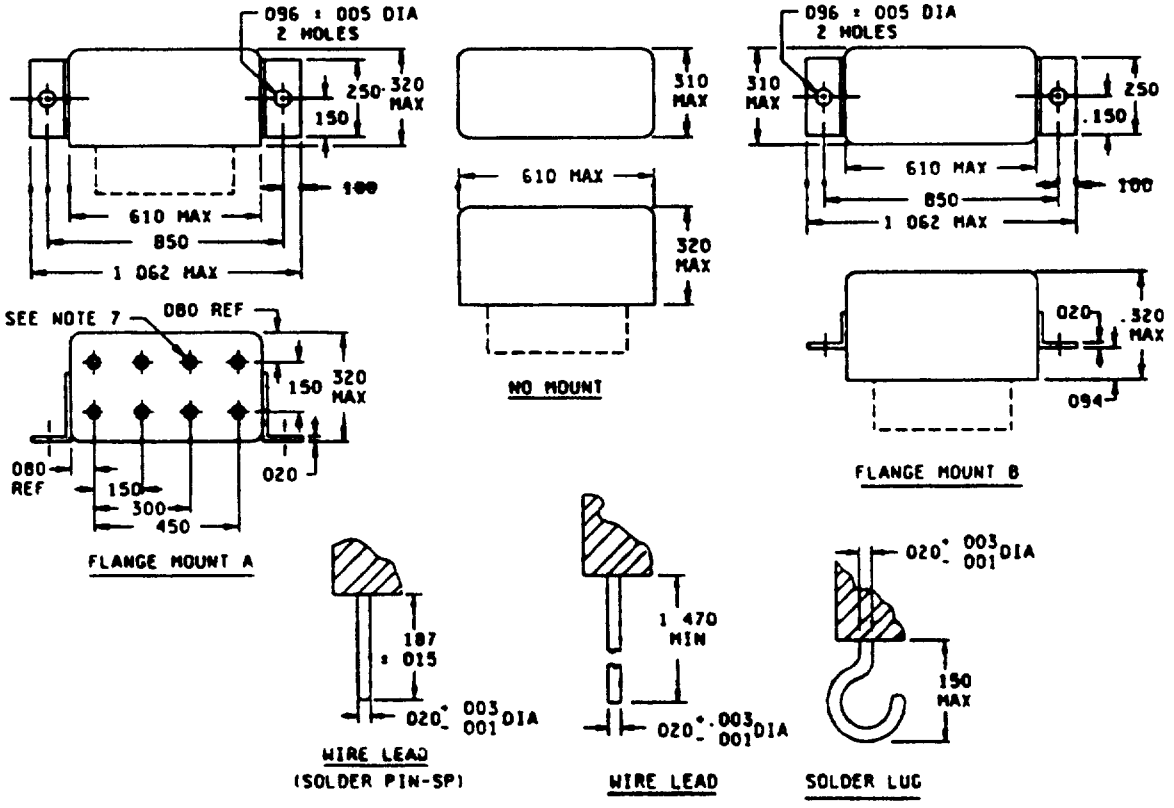
MIL-PRF 39016/37B
20 JULY 1988
SUPERSEDING
MIL-R-39016/37A
23 June 1981

PERFORMANCE SPECIFICATION SHEET

RELAYS ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 2 AMPERES (0 150-INCH TERMINAL SPACING) WITH INTERNAL DIODE FOR COIL TRANSIENT SUPPRESSION

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-R-39016

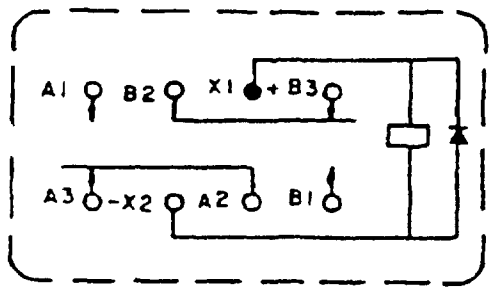


(9) FIGURE 1 Dimensions and configuration

(B) denotes changes

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CIRCUIT DIAGRAM
TERMINAL VIEW
DEENERGIZED POSITION
(SEE NOTES 6, 7, 8 AND 9)

Inches	mm	Inches	mm
001	0.03	.250	6.35
003	0.08	300	7.62
005	0.13	.310	7 87
.020	0.51	.320	8.13
.080	2.08	.450	11.43
.094	2.39	610	15.49
.096	2.44	850	21.59
100	2.54	1 062	26.97
.150	3.81	1.470	37.34
.187	4 75		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal locating dimensions shown are applicable to all type mounts
5. The shape of lug terminals is optional.
6. Coil symbol is optional per MIL-STD-1285.
(B) 7. X1 terminal shall be identified with contrasting bead.
(B) 8. Terminal markings A1 and A3 shall appear on the circuit diagram, other terminal markings are for reference only.
(B) 9. Relays shall have a plus sign marked on the circuit diagram as shown

(B) FIGURE 1 Dimensions and configuration - Continued

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

(B) Resistive: 2 amperes at 28 V dc (50,000 operations). 1 ampere at 28 V dc, 0.125 ampere at 115 V ac, (60 and 400 Hz) 0.5 ampere at 115 V ac, (60 and 400 Hz) with case not grounded. 400 Hz life test not required for qualification testing.

(B) Inductive: 0.3 ampere at 200 mH inductive at 28 V dc.

Lamp: 0.10 ampere at 28 V dc - Life test not required.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance and voltage drop:

(B) Initial. 0.050 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

(B) After life: 0.150 ohm maximum.

Low level.

During life: 33 ohms maximum.

(B) After life: 0.150 ohm maximum.

(B) Intermediate current:

During intermediate current: 1 ohm maximum.

(B) After intermediate current: 0.300 ohm maximum.

Contact bounce. 1.5 milliseconds (ms) maximum. (Applicable to failure rate level "L").

Contact stabilization time: 2.0 milliseconds (ms) maximum. (Applicable to failure rate levels "M", "P", and "R").

Overload (high level only): 4 amperes resistive at 28 V dc, 0.6 ampere inductive at 28 V dc. (ac not applicable.)

(B) COIL DATA: (See table I).

Operate time: 4.0 ms maximum over temperature range with rated coil voltage.

Release time: 6.0 ms maximum over temperature range from rated coil voltage.

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ELECTRICAL DATA.

Insulation resistance: 1/ 10,000 megohms minimum, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage: 1/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure, and between all contacts in the energized and deenergized positions	750	350 All terminals to case
Between case, frame, or enclosure and coil(s)	500	
Between all contacts and coil(s)	750	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	750	

DIODE CHARACTERISTICS:

Maximum negative transient. 1.0 V.

(B) Coil transient suppression. Applicable (Warning - reverse polarity on coil terminals will destroy the diode).

(B) Semiconductor in process screening: Applicable, the visual inspection method for semiconductors shall be in accordance with MIL-STD-750, Method 2074.

ENVIRONMENTAL DATA:

Temperature range. 65° to +125°C.

(B) Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

(B) Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts (Applicable to qualification and group C testing only).

(B) Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: 100 g.

PHYSICAL:

Terminals: See figure 1 and table I.

Terminal strength 1.5 ±0.2 lbs. (pull).

(B) Solderability. Applicable

Terminal twist test Applicable to wire leads

Dimensions and configuration See figure 1 and table I.

1/ Connect coil leads together to avoid damage to diode.

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Weight: 4.82 grams (0.17 ounce) maximum.

Identification marking (full): Applicable (see figure 1, note 8).

LIFE TEST REQUIREMENTS:

Ⓑ High level: 100,000 cycles.

Ⓑ Low level: 100,000 cycles plus 900,000 cycles mechanical life.

PART NUMBER: M39016/37 (dash number from table I and suffix letter designation failure rate level).

Ⓑ TABLE I. Dash number and characteristics. 1/

Dash number 2/				Coil voltage 3/ V dc		At 25°C				Over temp range		
Wire Lead (SP)	Solder lug	Wire Lead	Mount	Rated	Max	Coil resistance Ohms ±10%	Specified pickup value voltage (V dc)	Specified hold value voltage (V dc)	Specified dropout value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)
001	---	002	No mount	5	7	44	2.4	1.45	0.26	3.3	2.0	0.16
~~003~~	~~---~~	~~004~~	~~mount~~									
---	005	006	A	6	8	56	2.7	1.6	0.3	3.8	2.2	0.18
007	008	009	B									
010	---	011	No mount	9	12	140	4.4	2.6	0.5	6.0	3.6	0.3
012	---	013										
---	014	015	A	12	16	210	5.4	3.2	0.6	7.4	4.5	0.36
016	017	018	B									
019	---	020	No mount	18	24	650	9.5	5.6	1.0	12.8	7.7	0.6
021	---	022										
---	023	024	A	26.5	35	1350	13.5	8.1	1.5	18.0	10.8	0.9
025	026	027	B									

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 Vdc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles); L, 3.0, M, 1.0; P, 0.1; R, 0.01. Example, 001L - - - - 027R.

3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

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Ⓑ TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
24 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable.	M39016/37-023	24 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable.
	M39016/37-001 M39016/37-006 M39016/37-010 M39016/37-019	2 units, each part number, Qualification inspection table, group II.
	M39016/37-018	2 units, qualification inspection table, group II, and shock, vibration, acceleration, terminal strength, and seal.

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/38 may be used in addition to MIL-R-39016/37 data. Prior to performance of retention of qualification testing; the relay manufacturer shall preselect the sample size.

2/ The number of units required for qualification testing will be increased as required in group V, table II, MIL-R-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of retention of qualification inspection testing; the relay manufacturer shall preselect the sample size.

QUALIFICATION INSPECTION (reduced testing): See table III.

If the relays produced for MIL-R-39016/37 are similar in construction and design except for the steering diode and coil to the relays produced for MIL-R-39016/38, then reduced testing for qualification of MIL-R-39016/37 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/38 relays.

Ⓑ TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage - Group II of qualification inspection table.
1 unsealed sample unit - Internal examination.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Ⓑ Air Force - 85

Review activities
Army - AR, SL
Navy - AS, OS, SH
Ⓑ Air Force - 99
DLA - ES

User activities
Navy - MC
Ⓑ Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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